Hypermatter Reactor: System Design Description

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1. System overview

The hypermatter reactor is the station's main power plant. It supplies every zone substation through the power distribution system and is backed by the solar collector array.

The system is led by Reactor & Power Systems. It is delivered through 7 work packages with a combined budget of 79,008.44 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
HR-RCV	Reactor core vessel	SHV-HR-SPEC-0002	SHV-HR-DWG-0201	SHV-HR-BOM-0301
HR-CFG	Containment field generator	SHV-HR-SPEC-0102	SHV-HR-DWG-0202	SHV-HR-BOM-0302
HR-FIA	Fuel injector assembly	SHV-HR-SPEC-0103	SHV-HR-DWG-0203	SHV-HR-BOM-0303
HR-CMF	Coolant manifold	SHV-HR-SPEC-0104	SHV-HR-DWG-0204	SHV-HR-BOM-0304
HR-RMS	Reactor monitoring station	SHV-HR-SPEC-0105	SHV-HR-DWG-0205	SHV-HR-BOM-0305

3. Architecture

3.1 Reactor core vessel

The reactor core vessel is specified in SHV-HR-SPEC-0002. Its governing parameter is the core diameter, set at 1.85 km. The responsible engineer is Elara Drummond.

3.2 Containment field generator

The containment field generator is specified in SHV-HR-SPEC-0102. Its governing parameter is the field strength, set at 96.66 % nominal. The responsible engineer is Fallon Darrow.

3.3 Fuel injector assembly

The fuel injector assembly is specified in SHV-HR-SPEC-0103. Its governing parameter is the injection rate, set at 7.79 kg/s. The responsible engineer is Alira Aster.

3.4 Coolant manifold

The coolant manifold is specified in SHV-HR-SPEC-0104. Its governing parameter is the flow capacity, set at 964 m³/s. The responsible engineer is Nadia Mathers.

3.5 Reactor monitoring station

The reactor monitoring station is specified in SHV-HR-SPEC-0105. Its governing parameter is the sensor channels, set at 5,286 ea. The responsible engineer is Emrys Hune.

4. Interfaces

Document	Title
SHV-ICD-HR-TX	Interface Control Document: Hypermatter Reactor / Thermal Control & Heat Rejection
SHV-ICD-HR-PD	Interface Control Document: Hypermatter Reactor / Power Distribution
SHV-ICD-HR-CN	Interface Control Document: Hypermatter Reactor / Computer Core & Data Network

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-HR-01	Hypermatter Reactor: Design & analysis	Nils Errant	IN_PROGRESS	90.0%
WP-SHV-HR-02	Hypermatter Reactor: Fabricate & install reactor core vessel	Jessa Pell	IN_PROGRESS	74.6%
WP-SHV-HR-03	Hypermatter Reactor: Fabricate & install containment field generator	Jessa Pell	IN_PROGRESS	88.1%
WP-SHV-HR-04	Hypermatter Reactor: Fabricate & install fuel injector assembly	Edda Lancer	IN_PROGRESS	97.4%
WP-SHV-HR-05	Hypermatter Reactor: Fabricate & install coolant manifold	Edric Welles	COMPLETE	100.0%
WP-SHV-HR-06	Hypermatter Reactor: Fabricate & install reactor monitoring station	Ulric Kirwan	IN_PROGRESS	81.4%
WP-SHV-HR-07	Hypermatter Reactor: Integration & test	Edda Lancer	IN_PROGRESS	49.1%

6. Risks

- Supply of reactor core vessel assemblies depends on a single supplier.
- Supply of containment field generator assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.